

A provenance-sealed anomaly-score candidate catalogue of DESI DR1 spectra: selection, validation, and a descriptive taxonomy

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We publish a machine-flagged candidate catalogue of spectroscopically unusual objects in Data Release 1 of the Dark Energy Spectroscopic Instrument (DESI), together with the complete provenance chain needed to reproduce it. An archived convolutional autoencoder, bound by SHA-256 to a sealed run contract, assigns a reconstruction-error anomaly score S to every science-target spectrum in the public `iron` release; 27,547,223 unique `TARGETIDs` are scored. We show that a score cut alone does not select astrophysical objects: above $S = 3$, more than 99.5% of raw fibres in every score bin are sky or non-science fibres, and that fraction *rises* with score. Applying an explicit science-target provenance gate (`OBJTYPE = TGT`, `COADD_FIBERSTATUS = 0`, `TARGETID > 0`) before any score cut, and a pre-declared threshold rule, yields a released catalogue of 1,244 objects. We validate the catalogue against a published contract of sixteen machine-checkable requirements, and report two results that constrain its interpretation: within the science-target sample the anomaly score is uncorrelated with exposure quality and source brightness ($|\rho_s| < 0.1$ against median signal-to-noise, exposure time, `TSNR2` and r -band flux), so the tail is not a faintness artefact; and the score is driven almost entirely by the blue spectrograph arm ($\rho_s = +0.53$ against the b -camera residual, versus -0.04 and -0.14 for r and z). Cross-matching against SIMBAD and NED identifies counterparts for 569 objects (45.7%); the 675 unmatched objects are grouped into a descriptive taxonomy of 25 clusters and 8 families, which we show carries no material structure in the model’s own latent space and must therefore be read as a stratification of the candidate list rather than as physical classes. A known-object recovery benchmark against five reference classes recovers no class at the pre-declared bar. We name 32 follow-up targets in five rules-based tiers — including four anomaly-selected DESI-pipeline $z > 4$ quasar candidates, two of them at $z > 6$, with no catalogued counterpart — and list the open tests, led by the absent selection function, that any population-level use of this catalogue must clear first.

I. INTRODUCTION

Wide-field spectroscopic surveys produce far more spectra than can be inspected. Unsupervised outlier detection is the standard response [1], and it works: a model trained to reconstruct typical spectra flags the ones it cannot. What such a search usually does not publish is the part that decides whether its output means anything — the exact model, the exact code, the exact parent catalogue, the selection rule, and the reason the rule was chosen.

This paper publishes that part alongside the catalogue. Our primary deliverable is a scored candidate catalogue of DESI DR1 spectra *together with* a sealed provenance chain: the SHA-256 of the parent zcatalog, of the archived model, of the inference code, and of every intermediate product, bound under one run contract, with a re-runnable gate for the selection rule and a published validation contract of requirements a third party can check against the released files.

[PLACEHOLDER: one paragraph placing this among prior DESI/SDSS anomaly-detection catalogues, cited as prior work; see the release document’s neutral prior-work sentence]

Section II describes the data and the provenance chain,

Sec. III the model and the selection rule, Sec. IV the validation contract and what it establishes, Sec. V the catalogue, Sec. VI the descriptive taxonomy, Sec. VII the named follow-up set and the open tests, and Sec. IX what this catalogue is not.

II. DATA

A. DESI DR1

We use Data Release 1 of DESI [2, 3]: the public `iron` spectroscopic reduction, its redshift catalogue `zall-pix-iron.fits` (SHA-256 2d95ad99...), and the per-(survey, program, healpix) coadd spectra, streamed and scored in place rather than bulk-downloaded. Redshifts and spectral classifications are the pipeline’s own [4].

B. Provenance chain

Every stage binds the exact SHA-256 of its input and output under a single sealed run contract (generation `clean-rerun-6699d09ff886`):

- parent zcatalog 2d95ad99...;
- 36,634-group locator inventory, receipt-verified group by group;

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- archived model f5266ba4... and inference code 3e7efb24...;
- enrichment, cross-match, and taxonomy manifests, each naming its input hash and its output hash.

All eight links of that chain are verified in the validation contract (check V3, Table II), and all nine released artifacts re-hash to the values in the landing receipt (check V2).

III. METHOD

A. Anomaly score

Spectra are scored by an archived autoencoder (a $496 \rightarrow 128$ reconstruction network) frozen for this run; the score is the model’s reconstruction-error statistic, so a high score means a spectrum unlike the shapes the model learned to reproduce. The scoring pass covered 28,425,963 raw rows, deduplicated by last occurrence to 27,547,223 unique TARGETIDs. The enrichment stage’s independent MSE reproduction check passed for all 1,244 released rows with zero offenders (check V3b).

B. Science-target selection

A row enters the candidate pool only if OBJTYPE = TGT and COADD_FIBERSTATUS = 0 and TARGETID > 0, the last asserted rather than merely filtered. This gate is applied *before* any score cut and is enforced by a committed, re-runnable checker, [check_sample_provenance.py](#) (full path in the reproducibility manifest).

The gate is not a formality. Figure 1 shows the fraction of raw fibres that are sky or non-science as a function of score: it exceeds 99.5% in every bin above $S = 3$ and *increases* with score, from 99.53% in $[3, 4)$ to 99.97% in $[8, 10)$. A naive top-score selection from this scan returns instrument artefacts. Table I gives the raw and science-target counts on the threshold ladder.

C. Threshold rule

The release threshold follows a rule fixed before the science-only counts were inspected: from the grid $\{3, 4, 5, 6, 8, 10\}$, take the largest threshold whose science-only count is ≥ 300 ; if that count exceeds 1,500, step to the next grid point unless doing so drops the count below 300. The rule selects $S > 3$ with $n = 1,244$, reproduced independently in check V4.

TABLE I. Threshold ladder: all fibres versus science targets.

$S >$	All fibres	Science targets	Science fraction
3	320,418	1,244	0.39%
4	86,053	152	0.18%
5	52,188	44	0.08%
6	27,180	20	0.07%
8	3,810	2	0.05%
10	337	1	0.30%

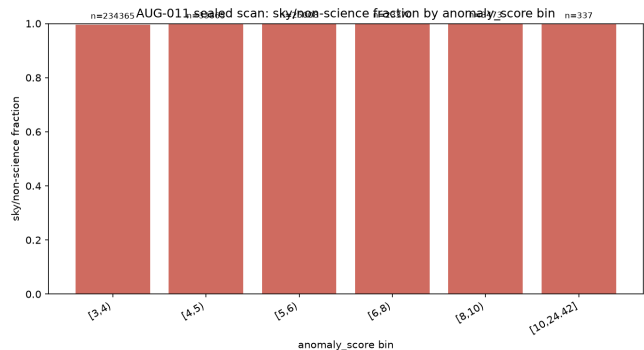


FIG. 1. Sky-or-non-science fibre fraction as a function of anomaly score, before the provenance gate. The fraction exceeds 99.5% in every bin above $S = 3$ and rises with score.

IV. VALIDATION

A. The validation contract

Table II lists the requirements this release is held to. Each is machine-checkable against the published files; the checker and its raw output are released with the catalogue. Requirements cover provenance and hash binding (V1–V3b), the selection rule (V4, V7, V9), internal consistency of the released tables (V5, V6, V8), and three tests of what the score measures (V10–V12).

B. The score is not an exposure-quality proxy

Within the science-target sample the anomaly score is essentially uncorrelated with every available measure of how well a spectrum was measured: Spearman $\rho_s = +0.058$ against median coadd signal-to-noise, -0.076 against coadd exposure time, -0.093 and -0.096 against TSNR2_{LRG} and TSNR2_{QSO}, and -0.016 against r -band flux (Fig. 2). At $n = 1,244$ several of these are formally significant and all are negligible in size; the two significant ones are negative, i.e. longer exposures score marginally *lower*. The claim this supports is narrow and negative: the $S > 3$ tail is not explained by faintness or short exposure. It is not evidence that the tail is astrophysically interesting.

TABLE II. Validation contract. Outcomes are re-run against the released artifacts; REPORTED marks a quantity the release must state rather than a threshold it must clear.

ID	Requirement	Outcome
V1	gates/check_sample_provenance.py exits 0 on the released sample tables (TARGETID $\neq$ 0; OBJTYPE/FIBERSTATUS where carried)	PASS
V2	every released artifact's SHA-256 equals the value in the landing receipt (PHASE3_V2_LANDING_2026-09-03.md)	PASS
V3	the manifest chain closes end to end: sample $\rightarrow$ enrichment $\rightarrow$ cross-match $\rightarrow$ taxonomy, each stage binding the exact SHA-256 of its input and output, under one run contract, with the model / inference-code / zcatalog hashes recorded	PASS
V3b	the enrichment stage's own MSE reproduction cross-check passed for every released row	PASS
V4	replaying the pre-declared threshold rule (largest grid point with science-only count $\geq$ 300, step up if $\geq$ 1,500) on the science-target counts selects the released threshold	PASS
V5	the cross-match output partitions the sample exactly: matched + unmatched = sample, disjoint, no duplicate TARGETIDs, and the enriched table covers the same TARGETIDs	PASS
V6	the taxonomy covers exactly the SIMBAD/NED-unmatched subset; 25 clusters roll up into 8 families; per-family counts sum to the unmatched count and match the per-object labels	PASS
V7	every released object exceeds the stated score threshold ($S \geq 3$)	PASS
V8	the AllWISE table covers every released object and its match count equals the released figure (74)	PASS
V9	the released sky-fraction-by-score validation supports the stated claim that every score bin above 3 is $\geq$ 99.5% sky-or-nonscience before the provenance gate	PASS
V10	within the science-target sample the anomaly score is not a re-statement of exposure quality or source brightness (—Spearman rho— $\geq$ 0.2 against S/N, exposure time, TSNR2 and r-band flux)	PASS
V11	the per-band residual decomposition identifies which camera arm drives the score (reported, not gated)	REPORTED
V12	the released taxonomy's family partition is tested for structure in the 128-dimensional latent space it was NOT built from (silhouette vs a label-permutation null, 50 draws)	STRUCTURED-BUT-WEAK
DEFECT-1	the enriched table's zcatalog columns (z, zwarn, spectype, subtype) are populated	FAIL
DEFECT-2	every derived residual-diagnostic column in the enriched table carries values	FAIL
DEFECT-3	the fraction of released objects carrying a DESI-reliable redshift (ZWARN == 0) is stated with the catalogue	REPORTED

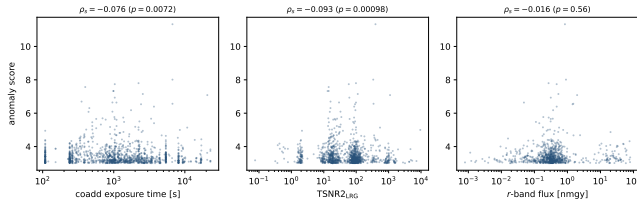


FIG. 2. Anomaly score against exposure time, TSNR2_{LRG}, and r -band flux for the 1,244 released objects, with Spearman coefficients.

C. The score is driven by the blue arm

Decomposing the reconstruction residual by spectrograph camera gives $\rho_s(S, r_B) = +0.527$ ($p \sim 10^{-89}$) against -0.037 for r_R and -0.142 for r_Z . Whatever the model responds to lives in the b camera. Whether that reflects blue-end sky-subtraction and calibration residuals or genuine spectral behaviour is the single most consequential open question about this catalogue, and it is testable on public data (Sec. VII).

D. Known-object recovery

Five reference classes with usable positions were cross-matched against the released sample at $1.5''$ (Table III). One broad-absorption-line quasar matches, at $4.2\times$ the base rate; no other class matches at all. Against the pre-declared bar — at least one class at $> 10\times$ enrichment with ≥ 5 matches — *no reference class is recovered*. This is the result that makes the present work a data release with a validated selection rather than a discovery paper, and we state it as such.

V. THE CATALOGUE

The released catalogue contains 1,244 objects, each carrying **TARGETID**, sky position, DESI survey and programme, pipeline redshift and classification, Legacy Survey photometry, per-band reconstruction residuals, the 128 latent activations, and the anomaly score, which ranges from 3.000 to 11.332.

Of the 1,244, 502 (40.4%) carry a reliable pipeline redshift ($\text{ZWARN} = 0$); the remaining 742 carry $\text{ZWARN} = 4$ (**SMALL_DELTA_CHI2**) and their redshifts must not be used as measurements. Every redshift-dependent statement in this paper is restricted to the $\text{ZWARN} = 0$ subset. Pipeline classifications are 1,153 **GALAXY**, 86 **QSO** and 5 **STAR**. AllWISE counterparts exist for 74 objects, with median $W1 - W2 = 0.51$.

SIMBAD [5] and NED cross-matching at $3''$ identifies counterparts for 569 objects (45.7%; 562 via NED, 38 via SIMBAD, some via both), leaving 675 (54.3%) unmatched. Unmatched does not mean new: it includes objects outside other surveys' footprints or below their detection limits. Notably, the three highest-scoring objects in the whole sample ($S = 11.33, 8.01, 7.80$) are all *matched*, i.e. already catalogued.

VI. DESCRIPTIVE TAXONOMY

The 675 unmatched objects are grouped by PCA $\rightarrow$ UMAP [6] $\rightarrow$ HDBSCAN [7] on (S, α, δ) into 25 clusters, rolled up by descriptor identity into 8 families (Table IV, Fig. 3). Family sizes sum to 675 exactly.

Because the clustering features are the score and the sky position, the families are in practice a score-tier $\times$ survey/programme stratification: 70.4% of objects lie in their family's dominant survey/programme cell, and no family is a compact sky region (the most concentrated still spans 52° in right ascension). We therefore tested the partition in the space the model actually encodes spectra into: its silhouette over the 128 latent dimensions is -0.016 against a 50-draw label-permutation null of -0.024 ± 0.004 (check V12). The partition is distinguishable from a random relabelling, but its absolute silhouette is negative — the families are *not* separated clusters in latent space. They are reported as descriptive

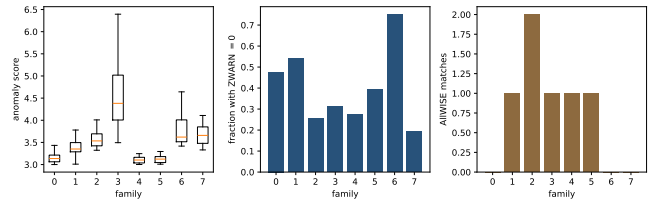


FIG. 3. Per-family anomaly-score distribution, reliable-redshift fraction, and AllWISE match count.

strata, carry descriptor labels only, and no family is given a physical class name.

Two families stand out for follow-up purposes. Family 3 (61 objects, single cluster) is the extreme-score family, with median $S = 4.38$, the largest median blue residual of any family, and the highest-scoring unmatched object in the catalogue ($S = 7.15$). Family 6 (36 objects) has the highest reliable-redshift fraction (75%) at an elevated score and no object above $z = 2$: well-measured, low-redshift, high-score, and therefore the family in which an instrumental or template explanation is most directly testable.

VII. FOLLOW-UP SET AND OPEN TESTS

A. Named targets

Table V names 32 objects in five rules-based tiers, all drawn from the unmatched subset. The rules are literature-motivated and replayable from the released files.

- **FT-A** (4 targets) — $\text{ZWARN} = 0$ and $z \geq 4$: four anomaly-selected DESI-pipeline $z > 4$ quasar candidates with no catalogued counterpart, two of them at $z > 6$. These are pipeline redshifts, not confirmations. The cheapest discriminating test costs nothing: a $z > 6$ quasar must be a g -dropout, so Legacy Survey DR10 grz photometry [11] at these coordinates refutes or supports each one on public data alone, before any spectroscopy.
- **FT-B** (1 target) — AllWISE $W1 - W2 \geq 0.8$, the standard mid-infrared AGN criterion [8, 9]. Next tests are public archives: NEOWISE-R light curve [10], ZTF optical light curve [13], archival X-ray coverage.
- **FT-C** (5 targets) — $\text{ZWARN} = 0$, $\Delta\chi^2 > 25$ and S in the top 5%: spectra that are simultaneously well measured and hardest to reconstruct, where a genuine spectral anomaly is most likely. Three of the five are in family 3.
- **FT-D** (0 targets) — unmatched point sources with a Gaia DR3 counterpart [12]. The rule returns nothing: the unmatched population is fainter than Gaia's

TABLE III. Known-object recovery benchmark, 1.5'' positional match.

Reference class	N_{ref} in footprint	N_{match}	Enrichment
Broad absorption line (BAL) quasars	5,285	1	4.2×
Roma-BZCAT blazars (5th ed.)	2,060	0	0.0×
Cataclysmic variables / WD binaries	580	0	0.0×
Lyman- α emitters	84	0	0.0×
Superluminous-SN host galaxies	27	0	0.0×

TABLE IV. Per-family evidence over the 675 SIMBAD/NED-unmatched objects. N_{cl} is the number of source clusters, r_B the median b -camera residual, N_{Z0} the number with $\text{ZWARN} = 0$, $\bar{z}_{Z0}$ their median redshift, and N_W the number with an ALLWISE counterpart.

Family	Tier	N	N_{cl}	Survey/prog.	Score med. (5–95%)	r_B	N_{Z0}	$\bar{z}_{Z0}$	$N_{z>2}$	N_{QSO}	N_W
0	low	302	12	main/dark	3.14 (3.02–3.38)	2.42	144	1.15	13	18	0
1	elevated	87	4	main/dark	3.35 (3.12–3.78)	2.53	47	1.20	8	10	1
2	elevated	71	2	sv3/bright	3.53 (3.35–4.06)	2.60	18	1.36	2	4	2
3	extreme	61	1	sv3/bright	4.38 (3.55–6.31)	2.98	19	1.19	1	2	1
4	low	44	2	sv3/bright	3.10 (3.01–3.23)	2.42	12	1.36	3	3	1
5	low	38	2	sv1/other	3.12 (3.01–3.28)	2.20	15	1.01	2	4	1
6	high	36	1	main/dark	3.62 (3.44–4.31)	2.81	27	1.09	0	1	0
7	high	36	1	sv3/bright	3.66 (3.37–4.01)	2.66	7	0.93	0	0	0

limit throughout. We retain the rule so that a deeper generation can re-run it.

- **FT-E** (25 targets) — the highest-score member of each cluster, so that a follow-up programme tests the taxonomy rather than one corner of it.

B. Open tests

OT-1 Selection function.: No injection-recovery experiment has been run for this model on this substrate. The catalogue therefore has no measured completeness or purity, and no population-level inference may be drawn from it. This is the gate any rate, abundance, or cosmological use must clear first. [PLACEHOLDER: run and report per-class injection recovery with the exact model and substrate named]

OT-2 Latent-space taxonomy.: Re-clustering on the 128 latent dimensions rather than on (S, α, δ) .

OT-3 Blue-arm diagnosis.: Compare the b -camera residual against airmass, lunar phase, and blue-end sky-subtraction residuals in the DR1 exposure tables, to separate calibration from astrophysics.

OT-4 Wider recovery benchmark.: Six of eleven candidate reference classes lacked usable positions and were not tested.

OT-5 Deeper scan.: A self-supervised model trained on the full science-target spectrum population, rather

than a 47k-spectrum autoencoder, is the realistic route to a confirmed anomalous class.

VIII. DATA AVAILABILITY AND REPRODUCIBILITY

The catalogue, its manifests, the validation checker and its output, the per-family evidence tables, and the follow-up set are released in the project repository under [results_2026-08-07/phase3_v2](#) and [flagship_assembly_2026-09-18](#) (both under [pipelines/p1_highz_tracers/](#)), with mirrors on Hugging Face and Backblaze B2 verified by checksum. The reproducibility manifest — external data sources and links, scripts, compute venue, cost, and wall-clock — is [REPRODUCIBILITY_MANIFEST.md](#). [PLACEHOLDER: Zenodo DOI for the public deposit]

Two defects in the released tables are recorded openly and must be repaired before public deposit: the enriched table's `z`, `ZWARN`, `SPECTYPE` and `SUBTYPE` columns are unpopulated (the authoritative values are in the sample table and are re-joined for every number in this paper), and `residual_kurtosis` is empty. The public release must ship the joined table. The released files also do not carry `OBJTYPE` and `COADD_FIBERSTATUS`, so the provenance gate cannot re-verify those two conditions from the published file alone; both columns should be carried in the deposit.

TABLE V. Named follow-up targets. S is the anomaly score. Redshifts are DESI pipeline values and are printed only where $\text{ZWARN} = 0$; a dash marks an object whose fit carries $\text{ZWARN} = 4$ (`SMALL_DELTA_CHI2`), for which the pipeline redshift is not a measurement. The FT-E tier spans the taxonomy by construction and therefore includes such objects.

Tier	TARGETID	R.A.	Dec.	S	z	ZWARN	SPECTYPE
FT-A	39627903570285763	247.89529	4.66991	3.496	6.0878	0	QSO
FT-A	39627607293039577	201.10137	-7.55334	3.232	6.0638	0	QSO
FT-A	39628216868014338	239.92839	18.01577	3.061	5.1938	0	QSO
FT-A	39627849358909321	252.47160	2.60073	3.033	4.3338	0	QSO
FT-B	103610381762578	223.91434	50.21732	3.105	–	4	GALAXY
FT-C	39627812453222421	212.59529	0.92095	7.151	1.0798	0	GALAXY
FT-C	39627812461610773	213.07783	0.96584	6.394	1.1868	0	GALAXY
FT-C	39627877817258319	149.92219	3.77396	6.180	1.1989	0	GALAXY
FT-C	39633001079899014	217.98717	36.00150	4.747	1.3533	0	GALAXY
FT-C	39633407088529181	97.49930	61.65664	4.642	1.3098	0	GALAXY
FT-E	39627812453222421	212.59529	0.92095	7.151	1.0798	0	GALAXY
FT-E	39633407088529181	97.49930	61.65664	4.642	1.3098	0	GALAXY
FT-E	39636182677589446	64.05184	-16.74091	4.321	0.0063	0	GALAXY
FT-E	1071051379310604	240.34378	42.49037	4.289	–	4	GALAXY
FT-E	39627748041295714	333.31949	-1.73961	4.252	1.4203	0	GALAXY
FT-E	39628291237221824	260.29164	21.13162	4.108	–	4	GALAXY
FT-E	39633396858619848	275.81585	60.77425	3.877	–	4	QSO
FT-E	1070259272417281	210.20609	6.12193	3.762	–	4	GALAXY
FT-E	1070185960177669	150.13551	2.91550	3.498	–	4	GALAXY
FT-E	39627903570285763	247.89529	4.66991	3.496	6.0878	0	QSO
FT-E	67803973419009	130.11975	18.65811	3.491	–	4	GALAXY
FT-E	39633418778053415	143.59822	62.82238	3.416	1.3659	0	GALAXY
FT-E	39628330143583760	240.61454	22.94852	3.400	3.1050	0	QSO
FT-E	39633432057219468	96.54153	63.88652	3.387	0.9304	0	GALAXY
FT-E	39633495382820413	102.70638	70.64374	3.334	–	4	GALAXY
FT-E	39633466035275117	293.95821	67.33832	3.330	1.4545	0	GALAXY
FT-E	39627817989705740	182.69435	1.34328	3.325	1.2954	0	GALAXY
FT-E	2394988512018434	203.98955	49.98956	3.295	–	4	GALAXY
FT-E	39633062929108448	182.23513	39.17878	3.279	0.2303	0	GALAXY
FT-E	39632946235181990	213.68195	33.19121	3.278	–	4	GALAXY
FT-E	1070090426515458	215.89907	-1.05994	3.247	–	4	GALAXY
FT-E	39627778101872111	325.03381	-0.42236	3.220	1.0189	0	GALAXY
FT-E	2394624320602116	188.48854	30.29515	3.216	–	4	GALAXY
FT-E	39633184245159076	270.34970	45.93852	3.194	1.4010	0	GALAXY
FT-E	39627654445405293	147.36232	-5.59562	3.158	–	4	GALAXY

IX. WHAT THIS CATALOGUE IS NOT

- It is not a discovery catalogue. No object here is reported as a confirmed unusual source; selection-pipeline output alone never promotes an object to notable.
- It is not a statistically characterised sample. Without OT-1 there is no completeness, no purity, and no selection function, so no population inference is available.
- Its families are not physical classes (Sec. VI).
- Its unmatched fraction is not a novelty fraction: it includes objects simply outside other surveys' footprints or detection limits.
- Its anomaly score is a distance from one model's learned reconstruction, not a physically motivated significance.

X. CONCLUSIONS

We publish 1,244 DESI DR1 science-target spectra selected by reconstruction-error anomaly score, with a sealed provenance chain, a published validation contract, a descriptive taxonomy of the unmatched subset, and a named follow-up set. Three results are worth carrying forward independently of the catalogue: a score cut without a provenance gate selects sky fibres, not objects, and the contamination worsens with score; within the science-target sample the score is not a proxy for exposure quality or brightness; and the score is driven by

the blue spectrograph arm, which localises both the next methodological test and the most plausible instrumental explanation. The known-object recovery benchmark does not clear its pre-declared bar, and we report the work as what it is — reproducible data infrastructure with a validated selection — rather than as a discovery.

ACKNOWLEDGMENTS

[PLACEHOLDER: DESI collaboration and facility acknowledgements; SIMBAD/NED/VizieR service acknowledgements; compute acknowledgement]

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